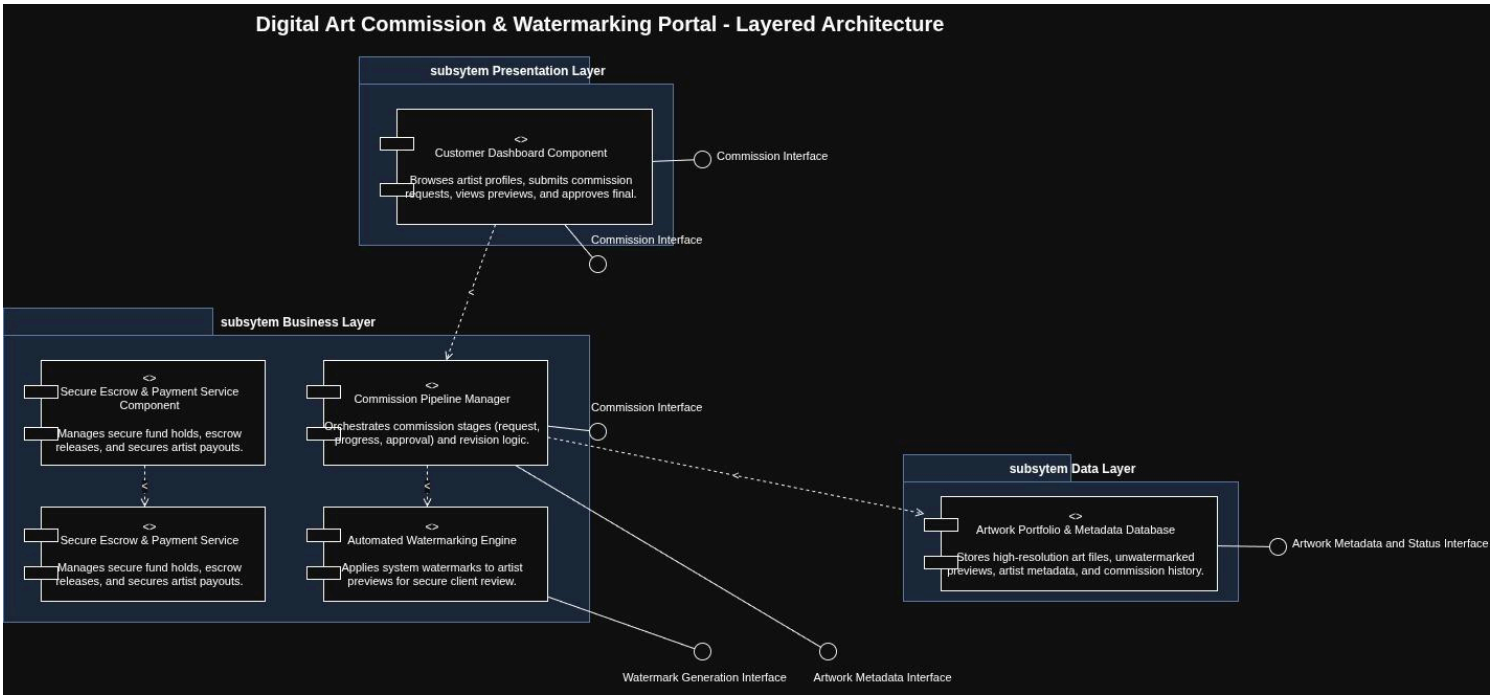
Lab 3: Architectural Justification System: Digital Art Commission & Watermarking Portal

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Architecture Selection: We chose Layered Architecture for the Digital Art Commission & Watermarking Portal System.

Architectural Choice: Layered Architecture.



Reasons for Selection The Layered Architecture pattern was selected based on the specific operational requirements of the system. The primary reasons include:

- **Separation of Concerns:** The architecture strictly isolates the user-facing dashboards (Presentation Layer) from the heavy backend processing required to dynamically render image watermarks (Business Layer). This prevents UI latency during heavy computational tasks.
- **Maintainability:** Business rules regarding commission approvals, revision logic, and payment routing are fully centralized in the business tier. This makes future platform

updates significantly easier to implement without affecting the user interface or database directly.

Security Advantage By completely isolating the Data Layer, the original unwatermarked, high-resolution art files cannot be accessed via a direct public URL or standard user interaction. They are locked securely in the Artwork Database Component and can only be unlocked and retrieved by the Business Layer once the secure escrow and payment triggers fire. This ensures artists' intellectual property is fully protected from theft prior to final payment.

Performance Benefit The Automated Watermarking Engine can be optimized, allocated more memory, or scaled independently within the Business Layer. This ensures the system can efficiently process large, high-resolution art files and render safe preview images for customers quickly, without slowing down the entire website's user interface or causing navigation lag for other users browsing the portal.